# TerraShift AI

Project Overview & Page Breakdown

## PROJECT IDEA

Monitoring large-scale land changes—such as **Saudi Vision 2030 projects (THE LINE, King Salman Park, Red Sea)**, deforestation, or urban growth—normally requires GIS specialists to manually download and compare gigabytes of satellite imagery by eye.

**TerraShift AI** automates this using **Deep Learning (Computer Vision)**. A user selects any location on Earth and two years (e.g., 2017 vs. 2024), and our **PyTorch Siamese U-Net** neural network compares the Before and After satellite images pixel-by-pixel to:

1. **Detect & Outline:** Automatically outline every new building, road, cleared forest, or water shift.
2. **Measure Area:** Calculate the exact area changed in km² and hectares.
3. **Forecast & Quantify:** Predict spatial growth up to 2030 and estimate the $CO_2$ and temperature impact.

## APPLICATION PAGES (7 PAGES)

| PAGE | ROUTE | WHAT IT DOES |
|---|---|---|
| **1. Overview** | / | Quick preview of the platform. Switch between 4 major case studies (*NEOM, Amazon, Dubai, Las Vegas*) and use the interactive Split Curtain or Magnifying Loupe to compare Before vs. After satellite imagery. |
| **2. Map Studio** | `/analyze` | Main interactive satellite map. Pick a preset or draw a custom box anywhere on Earth, select two years ($T_1$ and $T_2$), and run live AI change detection. Outlines all changed areas on the map and exports GeoJSON or a PDF report. |
| **3. Model Benchmarks** | `/benchmarks` | Compares our **PyTorch Siamese U-Net** against 3 traditional methods (*RGB Pixel Diff, NDVI Threshold, PCA K-Means*) on the same satellite pair using F1-Score, IoU, Precision, Recall, and neural network feature maps. |
| **4. 2030 Forecast** | `/timeline` | Shows a 5-year satellite filmstrip (2017–2024), identifies the exact year construction or deforestation accelerated the most (Breakpoint Year), and simulates future expansion through 2030 with CSV export. |
| **5. Pair Upload Lab** | `/lab` | Lets anyone test the AI on built-in benchmark patches (*LEVIR-CD*) or upload their own Before/After images from their computer. Outputs a probability heatmap, a True/False Positive confusion map, and an object table. |
| **6. Carbon & Heat** | `/carbon` | Translates land changes into environmental impact: calculates $CO_2$ emissions from tree loss, carbon removals from greening projects (like *Green Riyadh*), carbon credit value ($ USD), and surface temperature changes (°C). |
| **7. STAC Watchlist** | `/watchlist` | Catalog of 12 monitored hotspots (Vision 2030 sites, forests, hydrology). Users can filter by category, toggle 5-day satellite alert rules, inspect GeoJSON metadata, or open any site directly in the Map Studio. |

## TECH STACK

**Frontend**
Next.js 16, React 19, TypeScript, Tailwind CSS, MapLibre GL

**AI & Backend**
Python, FastAPI, PyTorch (Siamese ResNet-18 U-Net), NumPy, Shapely

**Data & Standards**
Sentinel-2 Imagery, Google Satellite Tiles, GeoJSON, STAC 1.0.0